

Unit III

3

HCI Models and Theories

3.1 User Profiles and Categorization of Users

Q.1 Describe briefly about user profiles.

Ans. :

- User profiling involves gaining an understanding and building a profile of the final users of the system in terms of age, gender, socio economic background, abilities, knowledge, skill sets, frequency, interest and any other relevant information.
- It included the process of establishing knowledge about the users to find out who users are, such as children, elderly, professional, male or female all the possibilities.

- It is important when your interface, website or object has multiple audiences, profiling will reveal potential clashes of interest. For instance, an educational website when viewed by parents, teachers and children will have to cater for each groups' different perspectives.
- The user is whoever is trying to get the job done using the technology. An appreciation of the way people's sensory systems (sight, hearing, touch) relay information is vital to design a first-class product.
- For example, display layouts should accommodate the fact that people can be side tracked by the smallest movement in the outer (peripheral) part of their visual fields, so only important areas should be specified by moving or blinking visuals. And of course,

people like designs that grab their attention. Designers must decide how to make products attractive without distracting users from their tasks.

Q.2 Explain the following :

- a) Designing for diversity
- b) Differences between user groups

Ans. :

a) Designing for diversity :

- In this world we have wide range of human abilities, backgrounds, motivations, personalities and intelligence presents major challenges for interactive system designers.

- They need to have an understanding of :

- Physical characteristics
- Cognitive and perceptual abilities
- Personality differences
- General abilities

- These are all characteristics, which apply to people in general, we shall also consider characteristics, which apply specifically to potential system users.

b) Differences between user groups :

- Consider the design of a check-out system for a large supermarket and the design of a counter system for a high-street building society.
- Talk to the person next to you and make a list of some of the differences between the groups of people who will use each system.
- List also how these differences could affect design decisions about each system.

Q.3 Observe skilled and novice operators in a familiar domain: for example, touch and 'hunt-and-peck' typists, expert and novice game players or expert and novice users of a computer application. What differences can you discern between their behaviors ?

[SPPU : Dec.-19, End Sem, Marks 5]

Ans. :

- The level of experiences between these two groups greatly differs from each other. Skilled or expert operators do better compared to the novice ones (of course).
- A novice operator needs to manage more information when learning a new skill whether a move or a technique for the computer application and consequently possesses less attention ability.
- Novices are more likely to experience anxiety during unfamiliar situations because they have lesser gone through compared to the experts.
- Skilled operators are better able to correct for extraneous influences on motor skill. For example, a skilled game player may have a greater chance of defeating the monster in the game compared to a novice player since he needs to master all the techniques needed to kill the hideous monster in the game.
- Touch typist's yields greater jobs done compared to a hunt-and-peck typists. Skilled operators can separate pertinent cues. They specifically know what information to attend to and are better able to focus in on these cues.
- Captivatingly, expert operators are better able to detect false cues as compared to novice operators. Better to say, expert or skilled operators has longer exposures to these technologies that is why they manage to do their pertinent tasks in an instant.

Q.4 List and explain categorization of users. Explain any two.

Ans. : There are a number of ways to categorize users :

- i) Primary and secondary users
- ii) 3 D framework
- iii) Occupational categories
- iv) Simple classification

Primary and secondary users :

- Primary (direct) user : The person, who actively uses the site. For example airline reservation clerk, Help desk staff.
- Secondary (indirect) user : The person being served by a primary user. For example Airline passenger and Customer, who called the support line.

Simple categorisation of users :

- These are Novice users.
- No syntactic knowledge, little semantic knowledge.
- May have little task knowledge.
- May have anxieties about computer use.
- Knowledgeable intermittent users.
- Can maintain semantic knowledge of task and computer concepts.
- Requirements for consistency of structures in interaction so that user inferences are supported.
- Good help facilities and documentation are required.

Frequent users

- Well trained in semantic and syntactic aspects.
- Rapid response time.
- Brief feedback.
- Abbreviated command sequences.
- Accelerators to move through dialogue sequences.

3.2 Goal and Task Hierarchy Model

Q.5 What is GOMS ? List and explain elements of GOMS.

Ans. :

- The GOMS is an acronym for goals, operators, methods and selection. GOMS is a task analysis technique.
- GOMS is family of user interface modeling techniques.

- The GOMS model has four components : goals, operators, methods and selection rules.

1. **Goals** - Tasks are deconstructed as a set of goals and subgoals. GOMS the goals are taken to represent a 'memory point' for the user.

2. **Operators** - Tasks can only be carried out by undertaking specific actions. Example : To decide which search engine to use.

3. **Methods** - It represent ways of achieving a goal. Example : drag mouse over field.

4. **Selection rules** - The method that the user chooses is determined by selection rules.

Q.6 Create a GOMS description of the task of photocopying a paper from a journal. Discuss the issue of closure in terms of your GOMS description.

[SPPU : May-18, End Sem, Marks 9]

Ans. :

- One possible GOMS description of the goal hierarchy for this task is given below. Answers will vary depending on assumptions about the photocopier used as the model for the exercise.

- In this example, we will assume that the article is to be copied one page at a time and that a cover over the imaging surface of the copier has to be in place before the actual copy can be made.

Goal : PHOTOCOPY-PAPER

Goal : LOCATE-ARTICLE

Goal : PHOTOCOPY-PAGE repeat until no more pages

[Select Goal : SELECT-PAGE --> CHOOSE-PAGE-TO-COPY]

Goal : ORIENT-PAGE

OPEN -COVER

POSITION-PAGE

CLOSE-COVER

PRESS-BUTTON

Goal : VERIFY-COPY

LOCATE-OUT-TRAY

EXAMINE-COPY

Goal : COLLECT-COPY

LOCATE-OUT-TRAY

REMOVE-COPY (outer goal satisfied!)

Goal : RETRIEVE-JOURNAL

OPEN-COVER

REMOVE-JOURNAL

CLOSE-COVER

Selection rules exist if a spoiled copy was printed. Consider the following :

Rule 1 : SELECT-PAGE if last page was copied successfully or start of article.

Note : The goal SELECT-PAGE is only valid if we are at the start of the article or the last copy was successful. If the last copy was spoiled the we must recopy the current page, so only a re-orientation would be required.

Goal : PHOTOCOPY-PAPER

Goal : LOCATE-ARTICLE

Goal : PHOTOCOPY-PAGE repeat until no more pages

[Select Goal : SELECT-PAGE --> CHOOSE-PAGE-TO-COPY]

Goal : ORIENT-PAGE

OPEN -COVER

POSITION-PAGE

CLOSE-COVER

PRESS-BUTTON

Goal : VERIFY-COPY

LOCATE-OUT-TRAY

EXAMINE-COPY

Goal : RETRIEVE-JOURNAL

OPEN-COVER

REMOVE-JOURNAL

CLOSE-COVER

Goal : COLLECT-COPY

LOCATE-OUT-TRAY

REMOVE-COPY (outer goal satisfied!)

- Closure to Outer Goal, must force user to collect copy last

Q.7 Explain advantages and disadvantages of GOMS.

Ans. : Advantages :

1. Easy to construct a simple GOMS model and saves time
2. Helps discover usability problems.
3. Gives several qualitative and quantitative measures.
4. Less work than usability study.

Disadvantages :

1. Only work for goal directed tasks.
2. Not for the novice user
3. Not ideal for leading edge technology systems
4. Not as easy as heuristics analysis, guidelines.

Q.8 Goals are accomplished by methods consisting of operators which are identified by selection rules. Illustrate this for following goals.

- i) To delete a sentence in a graphical text editor.
- ii) To close window in a graphical text editor.

 [SPPU : Dec.-17, End sem, Marks 9]

Ans. : i) to delete a sentence in a graphical text editor :

GOAL : DELETE SENTENCE IN A GRAPHICAL TEXT EDITOR

Method_for_goal: MENU-METHOD-DELETE SENTENCE

Step 1 : HIGHLIGHT SENTENCE

Step 2 : OPEN MENU

Step 3 : SELECT DELETE-COMMAND

Step 4 : Accomplish_goal MENU-METHOD-DELETE SENTENCE

Method_for_goal: DEL-KEY-METHOD-DELETE SENTENCE

Step 1 : POSITION-CURSOR AT END

Step 2 : PRESS DELETE FOR EACH LETTER

Step 3 : Accomplish_goal DEL-KEY-METHOD-DELETE SENTENCE

Selection_rules_for_goal : DELETE SENTENCE

If [long sentence] Then Accomplish_goal: MENU-METHOD-DELETE SENTENCE

If [short sentence] Then Accomplish_goal: DEL-KEY-METHOD-DELETE SENTENCE

ii) to close window in a graphical text editor

- When we have a window interface that can be closed in either of the two methods: by selecting the 'close' option from the file menu or by selecting the Ctrl key and the F4 key together.

- Model the task of "closing the window in a graphical text editor".

- Here, we have the high level goal of "close window" which can be achieved with either of the two methods: "use menu option" and "use Ctrl+F4 keys".

- This is unlike the previous example where we had only one method for each goal.

- We use the "Select" construct to model such situations (next slide)

Goal : Close window

[Select Goal : Use menu method

Operator : Move mouse to file menu

Operator : Pull down file menu

Operator : Click over close option

Goal : Use Ctrl+F4 method

Operator : Press Ctrl and F4 keys together]

- The select construct implies that "selection rules" are there to determine a method among the alternatives for a particular usage context.

- Example selection rules for the window closing task can be

- Rule 1: Select "use menu method" unless another rule applies
- Rule 2 : If the application is GAME, select "use Ctrl+F4 method"

- The rules state that, if the window appears as an interface for a game application, it should be closed using the Ctrl+F4 keys. Otherwise, it should be closed using the close menu option.

Q.9 What is cognitive complexity theory ?

Ans. :

- Cognitive complexity theory, introduced by Kieras and Polson, begins with the fundamental premises of goal breakdown from GOMS and enriches the model to offer added predictive power.
- CCT has two parallel descriptions : one of the user's goals and the other of the computer system (called the device in CCT). For the system grammar, CCT apply *generalized transition networks*, a form of *state transition network*.
- The description of the user's goals is based on a GOMS-like goal hierarchy, but is expressed primarily using *production rules*.
- The production rules are a sequence of rules :
if condition then action
- Where, *condition* is a statement about the contents of working memory. If the condition is true then the production rule is said to fire. An *action* may consist of one or more elementary actions, which may be either changes to the working memory or external actions such as keystrokes.

- As an example, consider an editing task using the UNIX vi text editor. The task is to insert a space where one has been missed out in the text. The fragment of the associated CCT production rules can be as below.

```
(SELECT-INSERT-SPACE
IF (AND (TEST-GOAL perform unit task)
(TEST-TEXT task is insert space)
(NOT (TEST-GOAL insert space))
(NOT (TEST-NOTE executing insert space)))
THEN ( (ADD-GOAL insert space)
(ADD-NOTE executing insert space)
(LOOK-TEXT task is at %LINE %COL) ))
(INSERT-SPACE-DONE
IF (AND (TEST-GOAL perform unit task)
(TEST-NOTE executing insert space)
(NOT (TEST-GOAL insert space)))
THEN ( (DELETE-NOTE executing insert space)
(DELETE-GOAL perform unit task)
(UNBIND %LINE %COL) ))
(INSERT-SPACE-1
IF (AND (TEST-GOAL insert space)
(NOT (TEST-GOAL move cursor))
(NOT (TEST-CURSOR %LINE %COL)))
THEN ( (ADD-GOAL move cursor to %LINE %COL) ))
(INSERT-SPACE-2
IF (AND (TEST-GOAL insert space)
(TEST-CURSOR %LINE %COL))
THEN ( (DO-KEYSTROKE ' ')
(DO-KEYSTROKE SPACE)
(DO-KEYSTROKE ESC)
(DELETE-GOAL insert space))
are,
• To see how these rules work, imagine that the user has just seen the typing mistake and thus the contents of working memory (w.m.) are,
(GOAL perform unit task)
(TEXT task is insert space)
(TEXT task is at 5 23)
(CURSOR 8 7)
```

• TEXT uses the text of the document that is being edited and CURSOR refers to the placing cursor on the screen. The position (5, 23) is the line and column of the typing error where the space is required. However, the present cursor location is at line 8 and column 7.

• So, the rule fires and its action is performed. This action has no external effect in terms of key-strokes, but adds extra information to working memory.

• The rules in CCT must not represent error-free performance. They can be utilized to explain error phenomena, even if they cannot predict them. The CCT rules are closely related to GOMS-like goal hierarchies: the rules may be generated from such a hierarchy or alternatively, it may analyze the production rules to obtain the goal tree.

• In fact, the CCT rules can characterize more difficult plans than the simple sequential hierarchies of GOMS. However, one should regard CCT as an engineering tool giving one a rough measure of learnability and difficulty combined with a detailed description of user behavior.

Q.10 Discuss the key differences between KLM and (CMN)GOMS.

 [SPPU : May-19, End sem, Marks 9]

Ans. :

- KLM is related to the GOMS model and can be thought of as a very low-level GOMS model where the method is given.
- The key difference between GOMS and KLM is how time is assigned to cognitive and perceptual operators when it comes to execution time predictions.
- Another major difference is that the goal hierarchy is explicit in GOMS while it was implicit in the KLM.
- The nature of unobservable operators is another important difference. KLM has a single M operator that precedes each cognitive unit of action. In contrast, GOMS assigns no time to such cognitive overhead.

• In (CMN)GOMS, a hierarchical cognitive (thought) process is assumed, as opposed to the linear thought process of KLM.

• (CMN)GOMS allows us to model the task and user actions in terms of four constructs (goals, operators, methods, selection rules)

• In KLM, only seven operators are defined. In (CMN)GOMS, the notion of operators is not restricted to those seven.

• KLM Contains several simplifying assumption. In (CMN) GOMS, slightly more specified that general GOMS and hierarchical goal structure and methods in program form.

3.3 Linguistic Model

Q.11 Write short note on linguistics models.

Ans. :

- Linguistic models represent the user-system grammar. Understanding the user's behaviour and cognitive difficulty based on analysis of language between user and system.
- Backus-Naur Form (BNF) and Task-Action Grammar (TAG) are used to represent this model.

Backus-Naur Form

- BNF can be used to define the syntax of a language.
- It is based on techniques developed for use with natural languages, but was specifically designed for use with computing languages.
- BNF defines a language in terms of terminal symbols, syntactic constructs and productions.
- Terminal symbols are the elementary symbols of the language, such as words and punctuation marks.
- In the case of computing languages, these may be variable-names, operators, reserved words, etc.
- Syntactic constructs (or non-terminal symbols) are phrases, sentences, etc.

- In the case of computing languages, these may be conditions, statements, programs, etc.

TAG :

- Task-action grammar (TAG) attempts to deal with some of these problems by including elements such as parametrized grammar rules to emphasize consistency and encoding the user's world knowledge.

- In BNF, three UNIX commands would be described as :

$\text{copy} ::= \text{cp} + \text{filename} + \text{filename} \mid \text{cp} + \text{filenames} + \text{directory}$

$\text{move} ::= \text{mv} + \text{filename} + \text{filename} \mid \text{mv} + \text{filenames} + \text{directory}$

$\text{link} ::= \text{ln} + \text{filename} + \text{filename} \mid \text{ln} + \text{filenames} + \text{directory}$

- No BNF measure could distinguish between this and a less consistent grammar in which

$\text{link} ::= \text{ln} + \text{filename} + \text{filename} \mid \text{ln} + \text{directory} + \text{filenames}$

Consistency of argument order made explicit using a parameter or semantic feature for file operations

- Feature possible values : Op = copy; move; link
- Rules

$\text{file-op[Op]} ::= \text{command[Op]} + \text{filename} + \text{filename} \mid$

$\text{command[Op]} + \text{filenames} + \text{directory}$

$\text{command[Op = copy]} ::= \text{cp}$

$\text{command[Op = move]} ::= \text{mv}$

$\text{command[Op = link]} ::= \text{ln}$

3.4 Physical and Device Models

Q.12 Explain keystroke-level model.

Ans. :

- The keystroke-level model (KLM) predicts how long it will take an expert user to accomplish a routine task without errors using an interactive computer system.

- The actions are termed keystroke level if they are at the level of actions like pressing keys, moving the mouse, pressing buttons.
- There is a standard set of operators for use in the KLM, whose execution times have been estimated from experimental data.
- The following is a step-by-step description of how to apply the KLM to estimate the execution time required by a specified interface design :

1. Choose one or more representative task scenarios.

2. Have the design specified to the point that keystroke-level actions can be listed for the specific task scenarios.

3. For each task scenario, figure out the best way to do the task or the way that you assume users will do it.

4. List the keystroke-level actions and the corresponding physical operators involved in doing the task.

5. If necessary, include operators for when the user must wait for the system to respond.

6. Insert mental operators for when user has to stop and think.

7. Look up the standard execution time to each operator.

8. Add up the execution times for the operators.

9. The total of the operator times is the estimated time to complete the task.

- The model decomposes the execution phase into five different physical motor operators, a mental operator and a system response operator :

○ **K** : Keystroking, actually striking keys, including shifts and other modifier keys.

○ **B** : Pressing a mouse button.

○ **P** : Pointing, moving the mouse (or similar device) at a target.

○ **H** : Homing, switching the hand between mouse and keyboard.

○ **D** : Drawing lines using the mouse.

- **M** : Mentally preparing for a physical action.
- **R** : System response which may be ignored if the user does not have to wait for it, as in copy typing.

Q.13 Explain limitations of keystroke-level model.

Ans. :

- It measures only one aspect of performance : time, which means execution time and not the time to acquire or learn a task.
- It considers only expert users. Generally, users differ regarding their knowledge and experience of different systems and tasks, motor skills and technical ability.
- It considers only routine unit tasks.
- The method has to be specified step by step.
- The execution of the method has to be error-free.
- The mental operator aggregates different mental operations and therefore cannot model a deeper representation of the user's mental operations. If this is crucial, a GOMS model has to be used.

Q.14 KLM (key - stroke - level) model predicts expert error - freetask completion time (human performance) with interactive computing systems. Total predicted time for a task is given by the equation. $t_{\text{EXECUTE}} = t_k + t_p + t_H + t_D + t_M + t_R$. What does each of the above timing represent ? Develop a KLM model and predict time for the completion of the task "Change font and style for the word "KLM" to bold, Arial" using mouse only.

Ans. [SPPU : May-18, Marks 9]

- Ans. :
- A task is broken into a series of subtasks. Total predicted time is the sum of the subtask times :

$$t_{\text{EXECUTE}} = t_k + t_p + t_H + t_D + t_M + t_R$$

- Operators :

K → keystroking, P → pointing, H → homing

D → drawing, M → mental prep, R → system response

Task : Change the font and style for word "KLM" to bold, Arial.

DECODE

• Operations :

Mouse Subtasks	KLM Operators	t_p (S)
Drag across text to select "KLM"	M P [2.5, 0.5]	0.686
Drag pointer to Bold button and click	M P [13, 1]	0.936
Move pointer to Font drop - down button and click	M P [3.3, 1]	0.588
Move pointer down list to Arial and click	M P [2.2, 1]	0.501
	$\sum t_p =$	2.71

• Prediction :

$$t_{\text{EXECUTE}} = 4 \times t_M + \sum t_p = 4 \times 1.35 + 2.71 = 8.11 \text{ seconds}$$

• Operations :

Keyboard Subtasks
Select text
Convert to boldface
Activate Format menu and enter Font sub - menu
Type a ("Arial" appears at top of list)
Select "Arial"

$$t_{\text{EXECUTE}} = 4 \times t_M + 12 \times t_k = 4 \times 1.35 + 12 \times 0.75 = 14.40 \text{ seconds}$$

Use "typing complex codes" ($t_k = 0.75$ s)

DECODE

3.5 Norman's 7 Stage Model

Q.15 What are the Norman's seven principles for transforming difficult tasks into simple ones?

Ans.: [SPPU: Aug-17, In Sem, May-19, End Sem, Marks-6]

Ans.:

- Donald Norman's seven stage of interaction are as follows :

1. Establishing the goal.
2. Forming the intention.
3. Specifying the action sequence.
4. Executing the action.
5. Perceiving the system state.
6. Interpreting the system state.
7. Evaluating the system state with respect to the goals and intentions

- **Stage 1 is Forming a Goal.** This is what you want. As an example, I might want a place that I can relax outside that will not get muddy and that I do not have to move my outdoor furniture around to mow.
- **Stage 2 is Forming the Intention.** This is what would satisfy the goal. A deck would satisfy my goal of place to relax outdoors that will not get muddy or be in the way of moving.
- **Stage 3 is Specifying an Action.** What do I have to do to achieve the intention ? I would need to build a deck to meet the requirement set in my goal.
- **Stage 4 is Executing the Action.** Here I would do the steps of the action. I would build the deck.
- **Stage 5 is Perceiving the State of the World.** Using the senses to gather information. My finished deck would be off the ground and have my outdoor furniture on it.

DECODE

3.6 Cognitive Architectures

Q.16 Explain Interacting Cognitive Subsystems (ICS) model.

Ans.:

- The Interacting Cognitive subsystems model is based on detailed cognitive experimentation which suggests that the human mind works by different subsystems passing information from one to another and copying it in the process.
- In this way, each subsystem has its own memory. Different systems operate with different coding, for instance, verbal, visual, auditory.
- There are higher order systems that translate these coding and integrate the information.
- The architecture of ICS is built up by the coordinated activity of nine smaller subsystems: five peripheral subsystems are in contact with the physical world and four are central, dealing with mental processes.
- Each subsystem has the same generic structure. A subsystem is described in terms of its typed inputs and outputs along with a memory store for holding typed information
- It has transformation functions for processing the input and producing the output and permanently stored information.
- Each of the nine subsystems is specialized for handling some aspect of external or internal processing.

DECODE

- An example of a central subsystem is one for the processing of propositional information, capturing the attributes and identities of entities and their relationships with each other.

3.7 Hierarchical Task Analysis (HTA)

Q.17 Consider the activity of making a telephone call. Record the actions in an HTA diagram or textually. Start off simply, assuming you know the number to dial, but then add more complicated situations: finding the number in an address book or what to do when the number is engaged.

[SPPU : May-18, End Sem, Marks 9]

Ans.:

- As with public pay phones :

0. make phone call
1. pick up receiver
2. dial number
3. wait for reply
4. talk
5. replace receiver

Plan 0 : 1 - 2 - 3

when answered - 4

when finished - 5

- We now add looking up the number. The form this takes depends on whether we find the number in an address book or a telephone directory.

- If both fail, say if the call is long distance to someone not in a local directory, the telephone operator must be consulted.

0. make phone call

1. find number

1.1 look in address book

1.2 look in phone directory

1.3 ask operator

1.3.1 pick-up receiver

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2. actually call

2.1 pick-up receiver

2.2 dial number

2.3 wait for reply

2.4 talk

2.5 replace receiver

Plan 0 : If number unknown - 1

When number found - 2

Plan 1 : if phoning friend - 1.1

If local call - 1.2

if 1.1 or 1.2 fail - 1.3

Plan 2 : 2.1 - 2.2 - 2.3

When answered - 2.4

When finished - 2.5

Finally, we add the case when the phone is engaged. The simplest way to do this is simply to change Plan 2.

Plan 2 : 2.1 - 2.2 - 2.3

If answered - 2.4 then when finished 2.5

If engaged - 2.5

However, looking at the second line it might suggest that we modify 2.4 to have two parts.

0. make phone call

...

2. actually call

2.4 successful call

2.4.1 talk

2.4.2 replace receiver

2.5 failed call

2.5.1 replace receiver

Plan 2 : 2.1 - 2.2 - 2.3

If answered - 2.4

If engaged - 2.5

Plan 2.4 : 2.4 then finished 2.5

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Q.18 A Hierarchical Task Analysis (HTA) provides an understanding of the tasks users need to perform to achieve certain goal. Perform HTA of the task - to cook food (rice). Illustrate using diagram.

Ans.: Refer Fig. Q.18.1.

[SPPU : Dec-17, End sem, Marks 9]

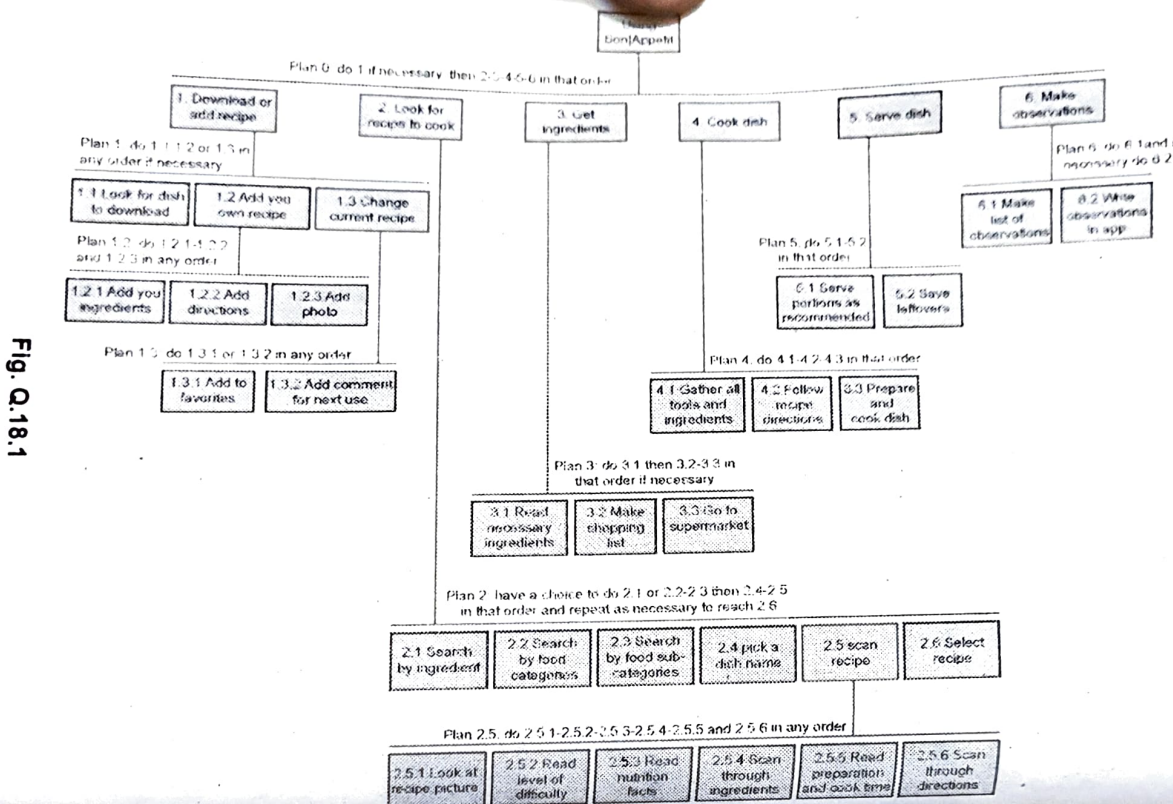


Fig. Q.18.1

Q.19 What is hierarchical task analysis? Explain with example.

Ans.:

- Task analysis is used mainly to investigate an existing situation. Hierarchical task analysis involves breaking a task down into subtasks, then sub-sub-tasks and so on.
- These are grouped as plans which specify how the tasks might be performed in practice.
- HTA focuses on physical and observable actions and includes looking at actions not related to software or an interaction device.
- Start with a user goal which is examined and the main tasks for achieving it are identified. Tasks are sub-divided into sub-tasks.
- Example of Hierarchical Task Analysis

0. In order to buy a DVD
 1. locate DVD
 2. add DVD to shopping basket
 3. enter payment details
 4. complete address
 5. confirm order
- plan 0 : If regular user do 1-2-3-4-5.

If new user do 1-2-3-4-5.

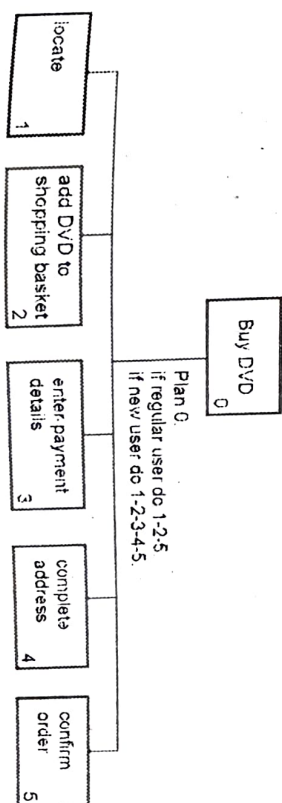


Fig. Q.19.1

Q.20 Hierarchical task analysis (HTA) is used to describe the interactions between a user and software system. Draw and explain HTA to online bus reservation system.

Ans. : [SPPU : Dec-18, End sem, Marks 9]

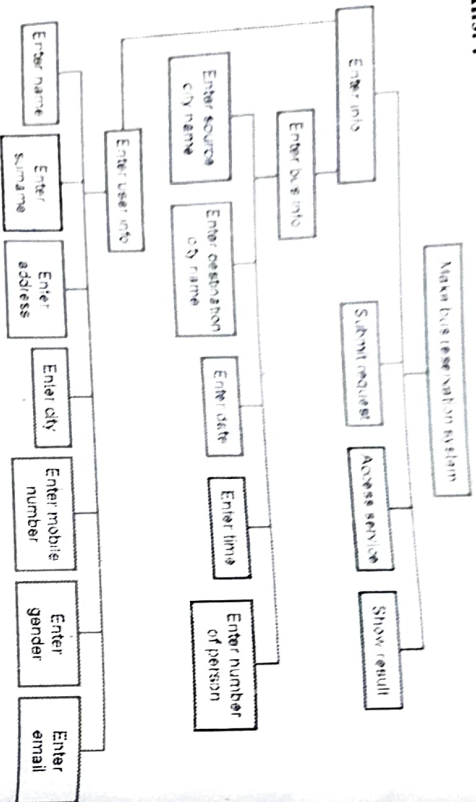


Fig. Q.20.1

Also refer Q.19.

Q.21 In order to clean the house

1. get the vacuum cleaner out
2. fix the appropriate attachment
3. clean the rooms
 - 3.1 clean the hall
 - 3.2 clean the living rooms
 - 3.3 clean the bedrooms
4. empty the dust bag
5. put the vacuum cleaner and attachments away

Plan 0 : do 1 - 2 - 3 - 5 in that order
When the dust bag gets full do 4

Plan 1 : do any of 3.1, 3.2 or 3.3 in any order

Depending on which rooms need cleaning.

For this HTA description of vacuum cleaning, present the same information in a diagrammatic form.

[SPPU : Dec-19, End Sem, Marks 9]

Ans. : HTA diagram for vacuum cleaning a house

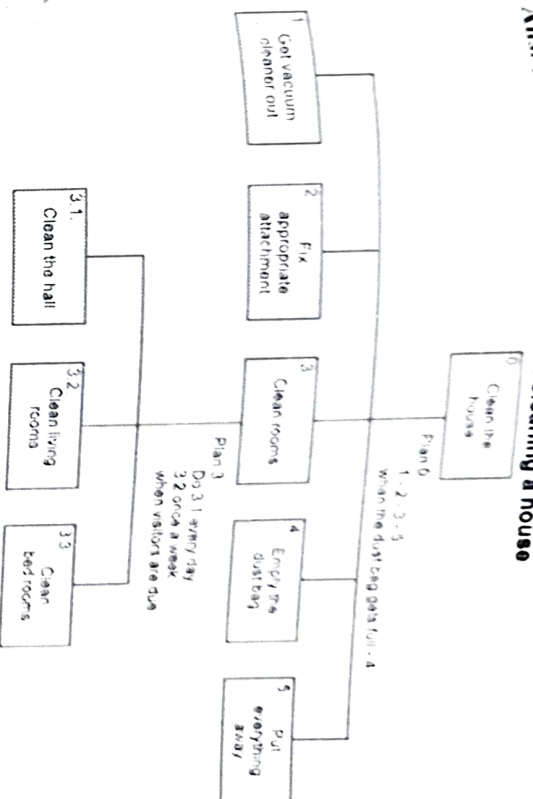


Fig. Q.21.1

3.8. Uses of Task Analysis

Q.22 How the task analysis help in requirements capture and systems design ?

Ans. :

- Task analysis in itself is not a form of requirements capture as it refers to the existing system, not the planned system and it includes many elements which are not part of the system.
- However, it makes a strong contribution toward the complete statement of requirements.
- The original statement of requirements given by a client will mention the new elements required and possibly refer to the existing system and its functionality.
- The task analysis of an existing system can help in two ways :
 1. The analyst can ask 'Which of the existing objects, tasks, etc., should be in the new system ?'.

- The formalized presentation of the existing state of affairs may help the client to clarify what the novel features are to be. It may be decided to automate whole tasks or roles or simply specific subtasks.
- As the high-level design of the system progresses, task analysis continues to play a role.

3.9 Diagrammatic Dialog Design Notations

Q.23 Draw a state chart diagram of a machine that dispenses bottles on inserting coins.

Ans. [SPPU : Dec-17, End Sem, Marks 9]

Ans. : State chart diagram for machine that dispenses bottles on inserting coins :

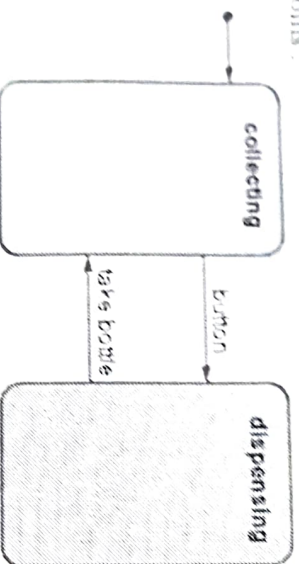


Fig. Q.23.1

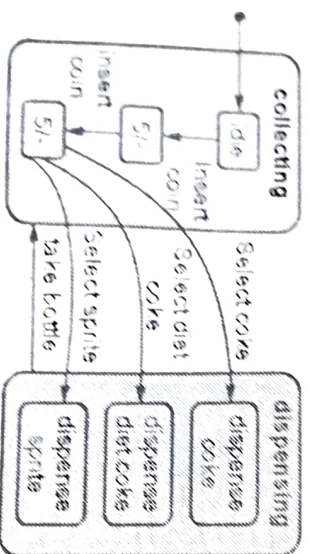


Fig. Q.23.2

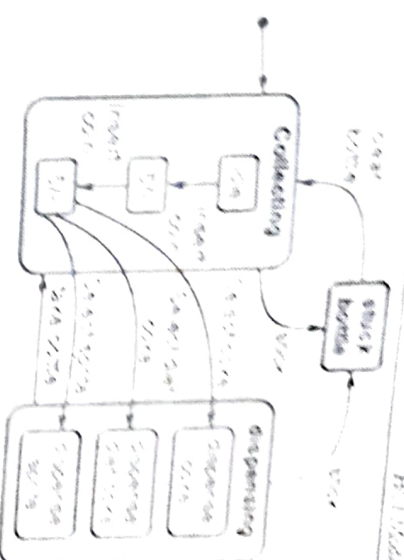


Fig. Q.23.3

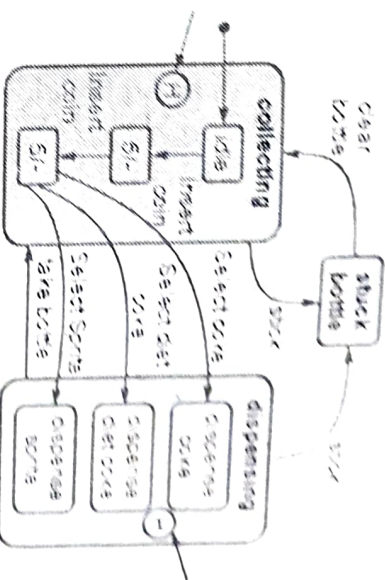


Fig. Q.23.4

Q.24 What is dialog ? How it is represent in computer language ?

Ans. :

- Dialog is a conversation between two or more parties.
- The dialog has a more specific meaning, namely the structure of the conversation between the user and the computer system.

Computer language at three levels :

- Lexical :** Lowest level and the shape of icons on the screen and the actual keys pressed. In human language, the sounds and spellings of words.
- Syntactic :** The order and structure of inputs and outputs. In human language, the grammar of sentence construction.

- **Semantic :** The meaning of the conversation in terms of its effect on the computer's internal data structures and/or the external world. In human language, the meaning ascribed by the different participants to the conversation.

Q.25 Explain State Transition Networks (STNs) ? How it is used for dialog description ?

Ans. :

- State Transition Network (STNs) are the most spontaneous, which knows that a dialog fundamentally denotes to a progression from one state of the system to the next.
- The syntax of an STN consists of the following two entities :
 1. **Circles :** A circle refers to a state of the system, which is branded by giving a name to the state.
 2. **Arcs :** The circles are connected with arcs that refers to the action event resulting in the transition from the state where the arc initiates, to the state where it ends.
- State transition networks have long been used for dialog description. Between the states are arrows, the transitions.
- These are labeled with the user actions that triggered the particular transition and the response the system makes.
- Fig. Q.25.1 shows state transition network for menu-driven drawing tool.

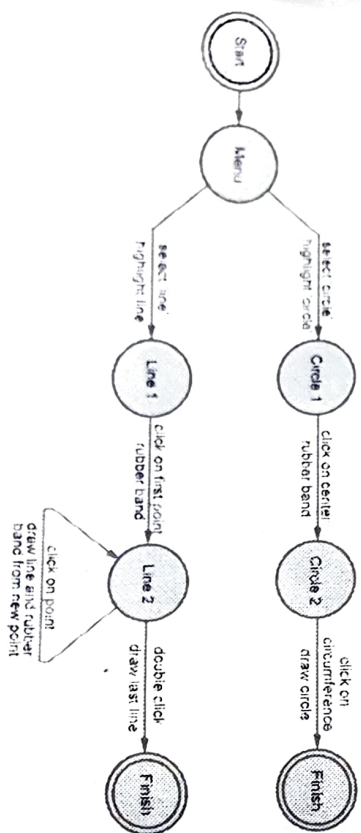


Fig. Q.25.1 : State transition network for menu-driven drawing tool

- We shall have a "start" state. From this "start" state, we shall go to a "menu" state, where we are shown the menu options.
- If we select the circle option, we go to a "circle" state. Otherwise, we select the "line" option and go to the "line" state.
- While at the "circle" state, we select a point as the circle center, which takes us to the "center" state. In the "center" state, we select the circle periphery and double click to indicate the end of input (the "finish" state). At this stage, the circle is displayed.
- While at the "line" state, we select a point as the beginning of the line. Then, we select another point to denote the last point on the line and transit to "point 2". At this stage, a line is displayed between the two points.
- We can select another point, while at "point 2" to draw another line segment between this point and the point last selected. We can actually repeat this as many times as we want, to draw line of arbitrary shape and size.
- When we perform a double click, it indicates the end of input and the dialog comes to the "finish" stage.

Q.26 Write short note on petri nets model.

Ans. :

- Petri net is a simple model of active behavior, which has four behavior elements such as – places, transitions, arcs and tokens.
- Petri nets provide a graphical explanation for easy understanding.
 1. **Place :** This element is used to symbolize passive elements of the reactive system. A place is represented by a circle.
 2. **Transition :** This element is used to symbolize active elements of the reactive system. Transitions are represented by squares/rectangles.
 3. **Arc :** This element is used to represent causal relations. Arc is represented by arrows.

4. **Token** : This element is subject to change. Tokens are represented by small filled circles.

• Tokens can play the following roles :

1. A physical object, for example a product, a part, a drug, a person.
2. An information object, for example a message, a signal, a report.
3. A collection of objects, for example a truck with products, a warehouse with parts or an address file.
4. An indicator of a state, for example the indicator of the state in which a process is or the state of an object.
5. An indicator of a condition: the presence of a token indicates whether a certain condition is fulfilled.

• Fig. Q.26.1 shows Petri nets model.

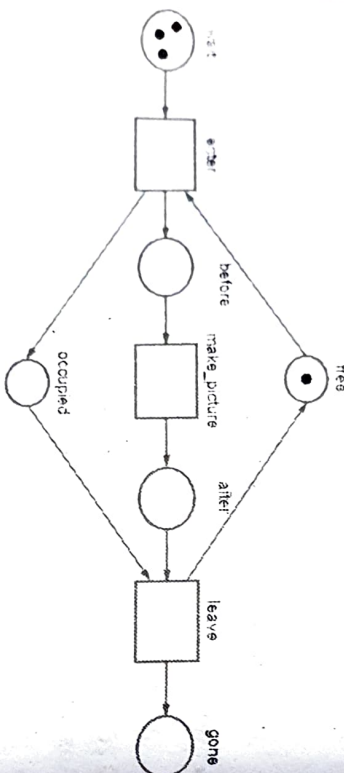


Fig. Q.26.1 : Petri nets model

• A transition can be used to represent things such as,

1. An event (e.g., starting an operation, the switching of a traffic light from red to green)
2. A transformation of an object, like adapting a product, updating a database or updating a document
3. A transport of an object : for example, transporting goods or sending a file

Q.27 Use a state diagram to describe the dialogue between the system and the user. Note any additional issues this raises about the system that need to be resolved in the design.

ES [SPPU : May-19, End Sem, Marks 9]

Ans. : Individual speech acts can contribute to a conversation. The basic structure of conversations can then be seen as instances of generic conversations. One example of such a generic structure is a Conversation for Action (CFA). This is shown as a state diagram in Fig. Q.27.1.

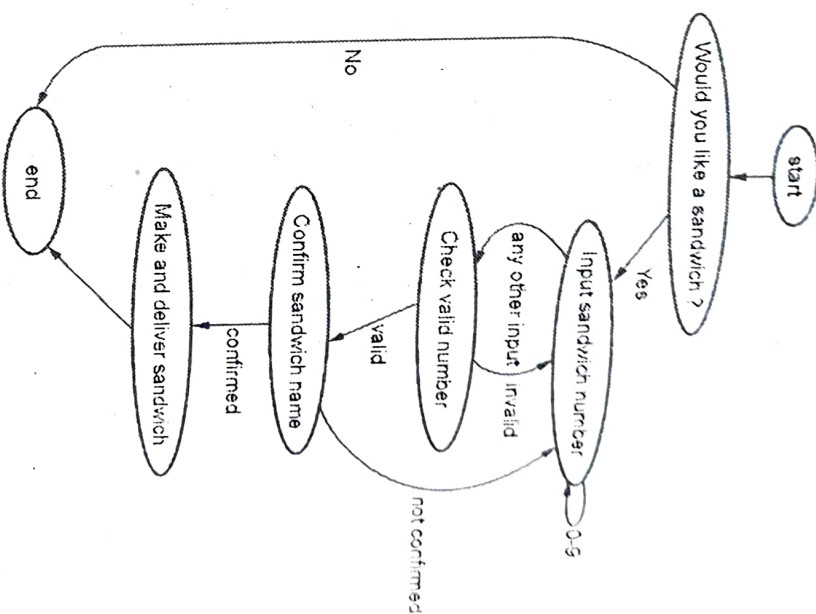


Fig. Q.27.1

END ...